



A  
Comprehensive Report on a Complex Engineering Problem  
Titled

AI-Based Resume Screening using Trees & Heaps

Data Structures: A9503

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# Complex Engineering Problem

## 1. Introduction

Modern recruitment processes involve handling thousands of resumes for a single job position, making manual screening time-consuming, inefficient, and prone to human bias. With the rapid growth of digital job platforms, organizations require intelligent systems to automate and optimize resume evaluation.

AI-based resume screening systems use data structures and algorithms to efficiently filter, rank, and shortlist candidates. Trees and heaps play a crucial role in organizing and prioritizing resume data. Trees help in structured decision-making and classification, while heaps enable efficient ranking of candidates based on scores.

This problem becomes complex due to the need for accuracy, fairness, scalability, and real-time processing while handling large volumes of unstructured data.

## 2. Problem Statement

Traditional resume screening methods fail to efficiently process large datasets and often result in delays and inconsistent candidate selection.

The core problem is to design an AI-based resume screening system using trees and heaps to:

- Efficiently filter and classify resumes
- Rank candidates based on multiple criteria (skills, experience, education)
- Reduce processing time for large datasets
- Ensure fair and unbiased selection

The complexity arises due to handling unstructured resume data, varying formats, and dynamic job requirements.

### **3. Objectives**

The objectives of this complex engineering problem are:

- To analyze inefficiencies in manual and basic automated resume screening
- To design a tree-based structure for classification and filtering of resumes
- To implement heap-based priority queues for ranking candidates
- To improve accuracy and speed of candidate selection
- To ensure scalability and fairness in recruitment systems

### **4. Constraints and Assumptions**

#### **Constraints**

- Large volume of resumes to process
- Limited computational resources in real-time systems
- Variability in resume formats (PDF, DOCX, etc.)
- Risk of algorithmic bias
- Data privacy and security concerns

#### **Assumptions**

- Resumes contain structured or semi-structured information
- Keywords and features (skills, education, experience) can be extracted
- Candidates can be scored based on predefined criteria
- Job requirements are clearly defined
- Input data is sufficiently clean for processing

### **5. Methodology**

The methodology adopted includes:

#### **1. Data Collection and Preprocessing**

- o Extract resume data using parsing techniques
- o Clean and structure the data (skills, experience, education)

#### **2. Feature Extraction**

- o Identify relevant keywords and attributes
- o Assign weights to different features

### **3. Tree-Based Classification**

- o Use decision trees to filter candidates based on eligibility criteria
- o Categorize resumes into shortlisted and rejected groups

### **4. Heap-Based Ranking**

- o Implement a max-heap to rank candidates based on scores
- o Retrieve top candidates efficiently

### **5. System Implementation**

- o Combine tree filtering with heap ranking
- o Optimize for time and space complexity

### **6. Evaluation**

- o Measure accuracy, processing time, and fairness
- o Compare with traditional screening methods

## **6. Results and Discussion**

The analysis reveals that:

- Tree structures efficiently filter resumes based on predefined rules
- Heap structures significantly reduce time required to find top candidates
- Automated systems improve consistency and reduce human bias
- Processing time is reduced compared to manual screening
- System performance depends on quality of feature extraction

However, challenges remain in handling unstructured data and ensuring unbiased decision-making. The results highlight that combining trees and heaps provides an efficient and scalable solution.

## **7. Recommendations and Conclusion**

### **Recommendations**

- Use advanced NLP techniques for better resume parsing
- Implement adaptive scoring models based on job roles
- Regularly update algorithms to reduce bias
- Ensure data privacy and ethical AI practices
- Integrate the system with real-time recruitment platforms

## **Conclusion**

AI-based resume screening using trees and heaps is a complex engineering problem involving challenges in data processing, algorithm design, and fairness. By leveraging efficient data structures, the system can significantly improve recruitment speed, accuracy, and scalability. Proper optimization and ethical considerations can make such systems highly effective in modern hiring processes.

## **8. References**

- Introduction to Algorithms – Cormen et al.
- Data Structures and Algorithms in C – Reema Thareja
- Research papers on AI in recruitment systems
- Online resources on Decision Trees and Heap Data Structures
- Tutorials on resume parsing and NLP techniques